



Model comparison, the Bayes Factor, and K2-18 b

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Model Comparison

- Frequentist approach: compare likelihoods, **focus on goodness-of-fit.**

Model parameters fixed. How likely are the observations under each model?

(χ^2 , likelihood ratio test)

Properly penalised: reduced χ^2 , Bayesian and Akaike Information Criteria

- Bayesian approach: compare posterior probabilities, marginal likelihoods, Leave-One-Out Cross-Validation, Widely Applicable Information Criterion.
Focus on out-of-sample predictive accuracy.

Model parameters random with distributions which are inferred from data and priors.

Automatically penalised!



Model Complexity in Bayesian Inference

- More model flexibility $\Rightarrow$ larger variety of predicted outcomes.
- BUT each outcome therefore predicted with lower probability!
(prob. density spread across larger space of possible outcomes)
- $\Rightarrow$ more flexible models are automatically penalised (Occam's Razor).
"of two competing theories, the simpler explanation of an entity is to be preferred."

One would prefer a model that gives a higher marginal likelihood, i.e., a higher likelihood of observing the data after integrating out the influence of the model parameter(s) (here: θ). A model will yield a high marginal likelihood if it makes a high proportion of good predictions (i.e., model 2 in Figure 13.2; the figure is adapted from Bishop 2006). Model predictions are normalized, that is, the total probability that models assign to different expected data patterns is the same for all models. Models that are too flexible (model 3 in Figure 13.2) will divide their prior predictive probability density across all of their predictions. Such models can predict many different outcomes. Thus, they likely can also predict the actually observed outcome. However, due to the normalization, they cannot predict it with high probability, because they also predict all kinds of other outcomes. This is true for both models with priors that are too wide or for models with too many parameters. Bayesian model comparison inherently penalizes more complex models, a principle known as *Occam's razor* (MacKay 2003). This principle favors simpler explanations when multiple viable explanations are possible.



Bayes' Theorem (Model Selection Version)

$$p(\vec{\theta}|D, \text{Model}) = \frac{\overbrace{p(D|\vec{\theta}, \text{Model})}^{\text{Likelihood}} \overbrace{p(\vec{\theta}|\text{Model})}^{\text{Prior}}}{\underbrace{p(D|\text{Model})}_{\text{Evidence/marginal likelihood}}}$$

$$p(D|\text{Model}) = \int d\vec{\theta} p(D|\vec{\theta}, \text{Model}) p(\vec{\theta}|\text{Model})$$



Bayes' Factor: Ratio of Evidences

$$B_{12} \equiv \frac{p(D|M_1)}{p(D|M_2)} = \frac{\int d\vec{\theta} p(D|\vec{\theta}, M_1) p(\vec{\theta}|M_1)}{\int d\vec{\theta} p(D|\vec{\theta}, M_2) p(\vec{\theta}|M_2)}$$

- No evidence ($B < 3$)
- Weak evidence ($3 < B < 12$)
- Moderate evidence ($12 < B < 150$)
- Strong evidence ($B > 150$)

- Frequentist version features only likelihood (parameters are fixed).
- More complex model predicts larger variety of outcomes
- BUT because of this, each outcome predicted with lower probability $\Rightarrow$ prob. density spread across larger space of possible outcomes.
- $\Rightarrow$ more complex models are automatically penalised (Occam's Razor).
- PROBLEM: multidimensional integrals!!!!



Nested Models

- If M_2 and M_1 are models such that M_1 is a special case of M_2 (e.g., quadratic model with quadratic term = 0 is a linear model), then M_1 is nested in M_2 .
- Easier to compute Bayes Factor because requires integration over only those extra variables.
- Codes specifically built to handle nested models (e.g., **dynesty**).

Demo!

To Colab!





Model comparison, the Bayes Factor, and K2-18 b

K2-18 b



NIRISS + NIRSpec data (2023)

Madhusudan+ 2023, ApJ Lett. 956, 1:

“[S]trong detections of CH_4 (5 sigma) and CO_2 (3 sigma).”

DMS detection was 1 sigma (i.e., no evidence), but press releases claimed evidence of this biosignature.

Schmidt+ 2025, arXiv:2501.18477

“[T]hese... were derived using a single data reduction and retrieval modeling framework, which did not allow for standard robustness tests. We confirm... CH_4 (**4 sigma**),... but **we find no statistically significant or reliable evidence for CO_2 or DMS.**”

<https://bsky.app/profile/distantworlds.space/post/3k7akv3ck6m2z>



NIRISS + NIRSpec data (2023)

Robustness testing crucial because:

- Data reduction introduces subtle systematics; using just one method risks undetected artifacts.
- Retrieval modeling frameworks can give different results depending on assumptions (e.g. priors, opacities, line lists, cloud treatments).
- Without testing alternative models, one cannot know if the result is unique or merely one of many statistically plausible explanations.

The same authors have shown in another paper (Constantinou+ 2023 ApJ Lett. 943, 10) that, depending on the pipeline, the derived abundances vary over an order of magnitude!



NIRISS + NIRSpec data (2023)

Therefore, the claims are preliminary and potentially unreliable, because:

- The pipeline might have introduced unrecognized biases,
- The retrieval framework might have converged on a local optimum or overfit,
- No competing model families or alternative data treatments were explored.
- This is a caution that the evidentiary basis for these dramatic interpretations is not secure without replication or cross-validation.



MIRI data (2025)

Madhusudhan et al. ApJ Lett. 983, 40

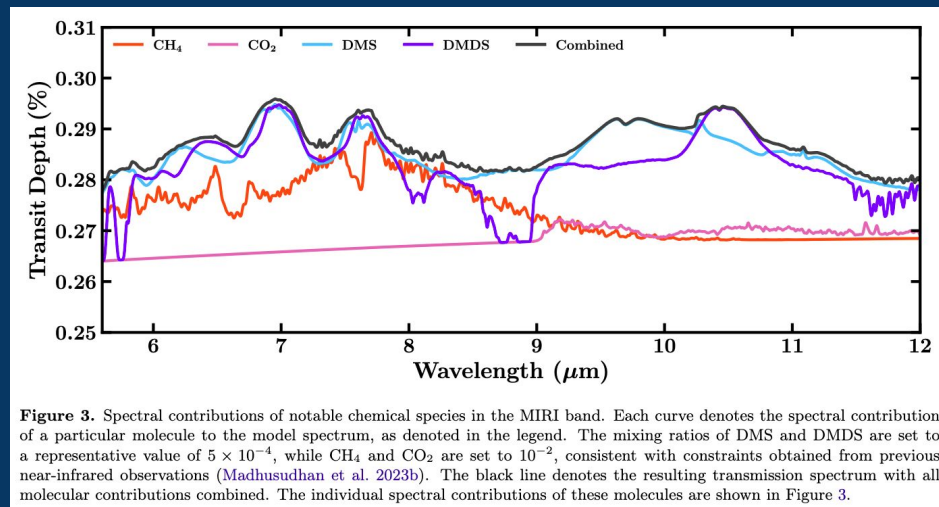
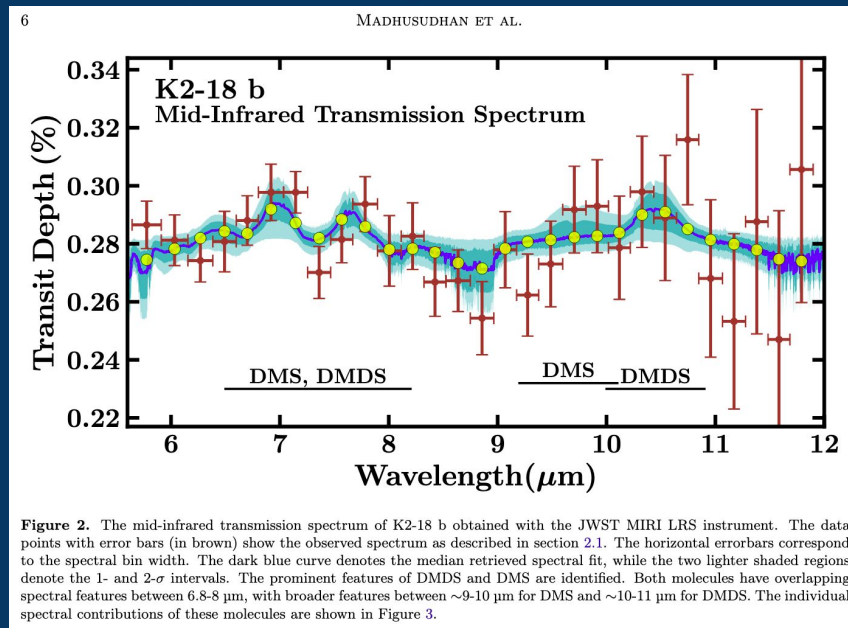
“The spectrum shows distinct features and is inconsistent with a featureless spectrum at 3.4σ significance compared to our canonical model. We find that the spectrum cannot be explained by most molecules predicted for K2-18 b, with the exception of DMS and dimethyl disulfide (DMDS), also a potential biosignature gas. We report new independent evidence for DMS and/or DMDS in the atmosphere at 3σ significance...”

The paper tests two different pipelines and discusses their retrieval setup in a lot of detail!



MIRI data (2025)

Spectrum only has 29 data points



“Canonical” model has 16 free parameters
(4 molecules)

“Flat line rejected at 3.4 sigma significance”

“Maximal” model has 32 free parameters
(20 molecules)



Model comparison, the Bayes Factor, and K2-18 b

MIRI data (2025)

<https://bsky.app/profile/distantworlds.space/post/3lmzihugafk2x> [Ryan MacDonald]

<https://bsky.app/profile/astrojake.bsky.social/post/3lnhm2gj3h224> [Jake Taylor]

<https://arxiv.org/abs/2504.15916>



MIRI data (2025)

Model comparison: “canonical” (16 parameters) vs flat line (1 parameter)

- flat line model is not subset of canonical model, so not a nested analysis.
⇒ Bayesian evidence calculation needs to take prior ranges into account.
- “canonical” models only considers contributions from DMS/DMDS, thus boosting the detection significance.
- the “maximal” model without DMS/DMDS can only reject the flat line model at < 2 sigma.

Taylor, J arXiv:2504.15916 – a truly nested analysis (flat line vs Gaussians)

$$\delta_\lambda(A, \mu, \sigma_m, c) = A \exp\left(-\frac{(\lambda - \mu)^2}{2\sigma_m^2}\right) + c \quad (1)$$

where δ_λ is the model transit depth, and μ , σ_m , and A are the centre, standard deviation, and amplitude of the Gaussian, respectively, and c is a vertical offset. Given that the spectral shapes of DMS and DMSs are Gaussian-like (see Figure 3 of N. Madhusudhan et al. (2025)), this is a reasonable approximation. I fit for the flat line using a model of the form:

$$\delta_\lambda(c) = c. \quad (2)$$

Four models: (a) flat line (b) Gaussian emission (c) Gaussian absorption (d) Gaussian emission + absorption Plus (b), (c), and (d) with feature centre fixed (7 and 8.8 μm).

Result: NO EVIDENCE for a Gaussian (ln B < 1)
In all cases except emission + absorption with fixed locations (WEAK EVIDENCE).

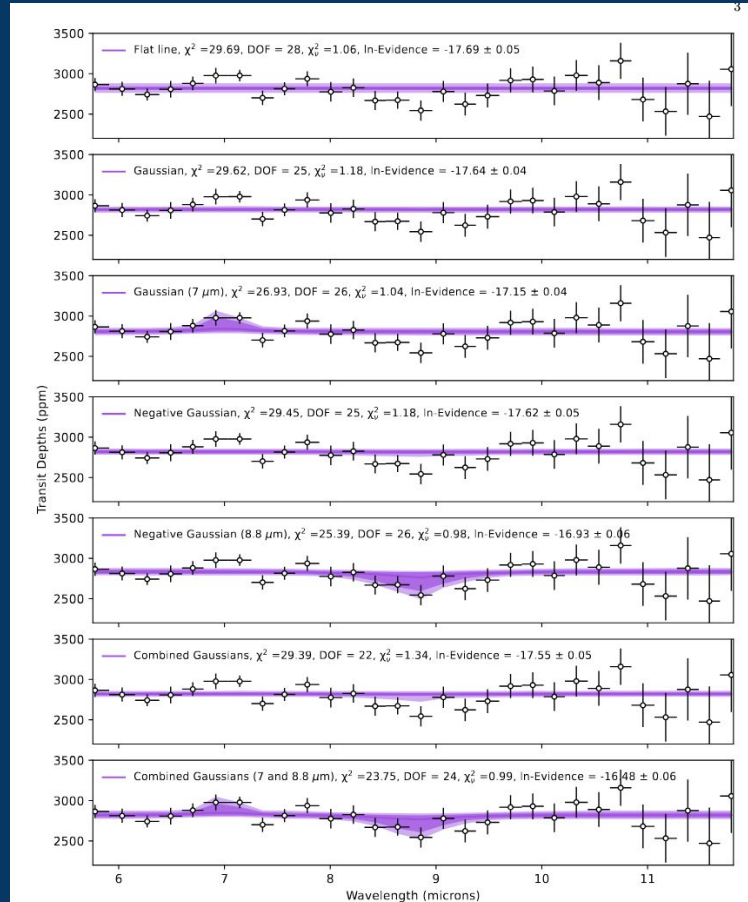


Figure 1. Model fits to the MIRI/LRS observations of K2-18b from (N. Madhusudhan et al. 2025). The median, 1- σ and 2- σ credible region envelopes for each model fit are overlaid (purple curve and shading). The χ^2 , degrees of freedom, reduced χ^2 , and the ln Bayesian evidence values are annotated.

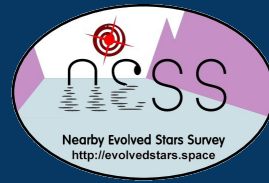


James Davies

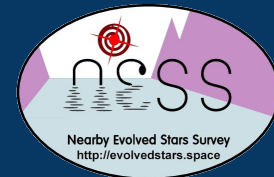
If you've ever tasted your friends' homebrew beer and it tasted like cooked corn, you were tasting DMS in the wort not being boiled off properly during production.

Clearly, the evidence shows that K2-18b space aliens are making lots of beer and apparently doing it well.

Null hypothesis: inhabitants of K2-18b are boring and sober.



Variables in the Nearby Evolved Stars Survey



AGB studies in the Milky Way

- **Foreground extinction** → **uncertain distances** → uncertain luminosities/dust-production rates.

Pulsation, convection, dust obscuration on the AGB → imprecise parallaxes.

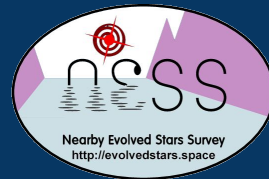
IRC +10216 (CW Leo, ~90 pc) Gaia DR3 parallax uncertainty: ~40%

Mira (omi Cet, ~90 pc) Hipparcos parallax uncertainty: ~10%

- Workaround: study
 - **compact populations** like Bulge AGB stars (e.g., Schultheis+ 2017, Olofsson+ 2022)
 - **small/biased samples** (e.g., Kemper+ 2003, Teyssier+ 2006, McDonald+ 2018), or limited scope (e.g., only carbon stars; Oloffson+ 1993),
 - **larger samples with relatively low detection rates** (e.g., Olofsson+ 2002, Loup+ 1993). **Stochasticity** or large spread in properties (e.g., gas:dust ratios, Knapp+ 1985).
- Need a **volume-complete**, statistically representative sample in the Solar Neighbourhood (e.g., Jura & Kleinmann 1989, 1990)



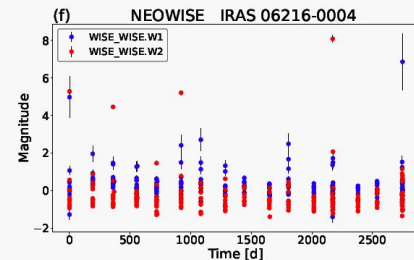
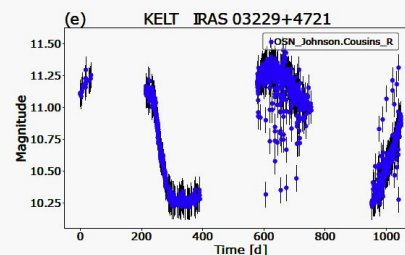
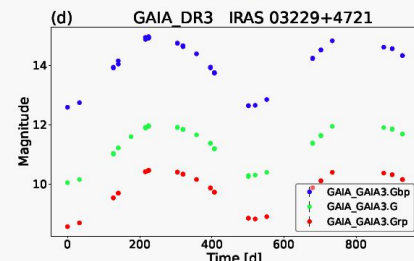
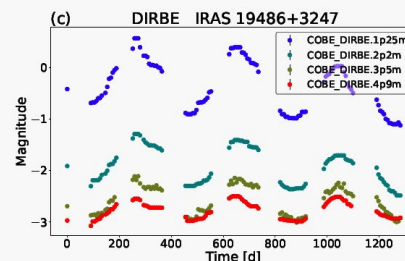
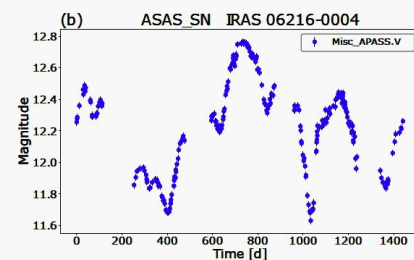
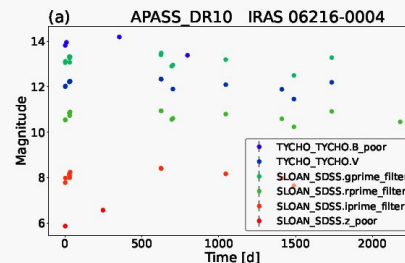
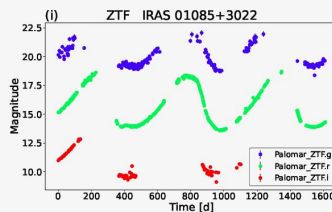
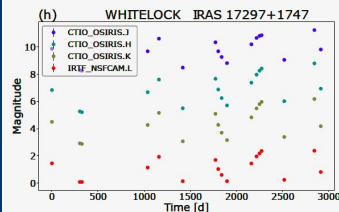
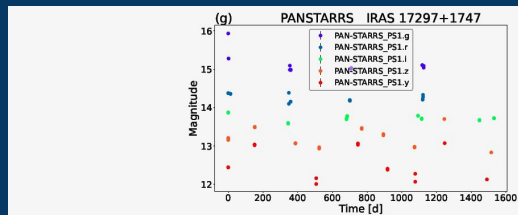
The Nearby Evolved Stars Survey

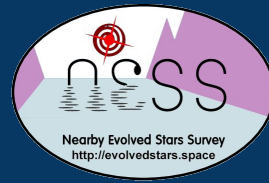


- Volume-complete (< 3 kpc) survey of ~ 850 evolved candidates selected based on IRAS colours. Sample most evolved objects **as well as early phases of mass loss**.
- Combine gas and dust data to study mass loss [history].
- Overview paper: Scicluna+ (2022), JCMT data
In prep: McDonald, Srinivasan+ (catalogue), Wallström+ (JCMT heterodyne), Jeste+ (APEX), Amada+ (NRO), Trejo+ (ALMA-ACA) [Hope you were awake during Alfonso's talk...]
Observations in progress: IRAM, LMT, VLA
- **Lots of archival data over entire relevant range of wavelengths!**

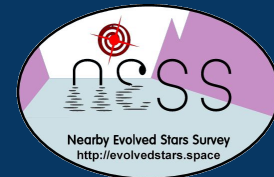
The light-curve dataset

- Nine surveys, 29 photometric bands.
- Can study variability as a function of wavelength!
- At least one band of data for 712/852 targets.
- Fully reproducible! Python scripts (`wget` or `ADQL`), fits, and data will be available with paper.
- A total of >6700 light curves.





Fits to NESS variables with the PGMUVI code

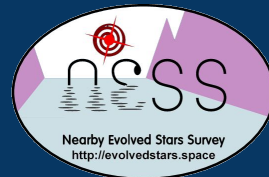


Motivation

- Data from many instruments/telescopes/surveys: wide range of systematics/noise properties. Combine data from multiple wavelengths for better understanding of physical process driving variability.
- Wavelength dependence could be in amplitude (e.g. eclipsing binaries), in phase (e.g. pulsating stars), or even period (e.g. multiperiodic systems).
- Gaussian Process Regression can model such data!
- Wavelength dependence being tested, results soon...



PGMUVI: GPR for time series



- Model: light curve data is drawn from a multivariate Gaussian with a covariance (kernel) describing the dependence between any two points in the light curve.

$$[f(t_1), f(t_2), \dots, f(t_N)]^T = \mathbf{f}(\lambda, \mathbf{t}) \sim N(\boldsymbol{\mu}(\mathbf{t}), \mathbf{K}(\mathbf{t}, \mathbf{t}'))$$

- Python Gaussian process for MUltiwavelength Inference (PGMUVI)

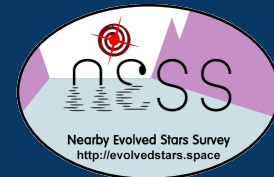
Scicluna+, JOSS, submitted <https://ui.adsabs.harvard.edu/abs/2023arXiv230800132S>

Code: <https://github.com/ICSM/pgmuvi>, Doc: <https://pgmuvi.readthedocs.io>

- PGMUVI uses a spectral mixture kernel (Wilson+ 2013) to compute a Fourier-space fit.



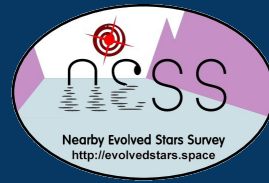
PGMUVI fit procedure



- Use a [multiband] Lomb-Scargle periodogram fit to obtain initial guesses for periods.
- Perform a 2-component fit (two periods) to the observed light curve at a given wavelength.
- TBD: fit light curves at multiple wavelengths simultaneously.
Code works, visualisation/interpretation of results TBD.
- For each source, compute P0 and P1 for each light curve.
Compute median periods from these estimates.



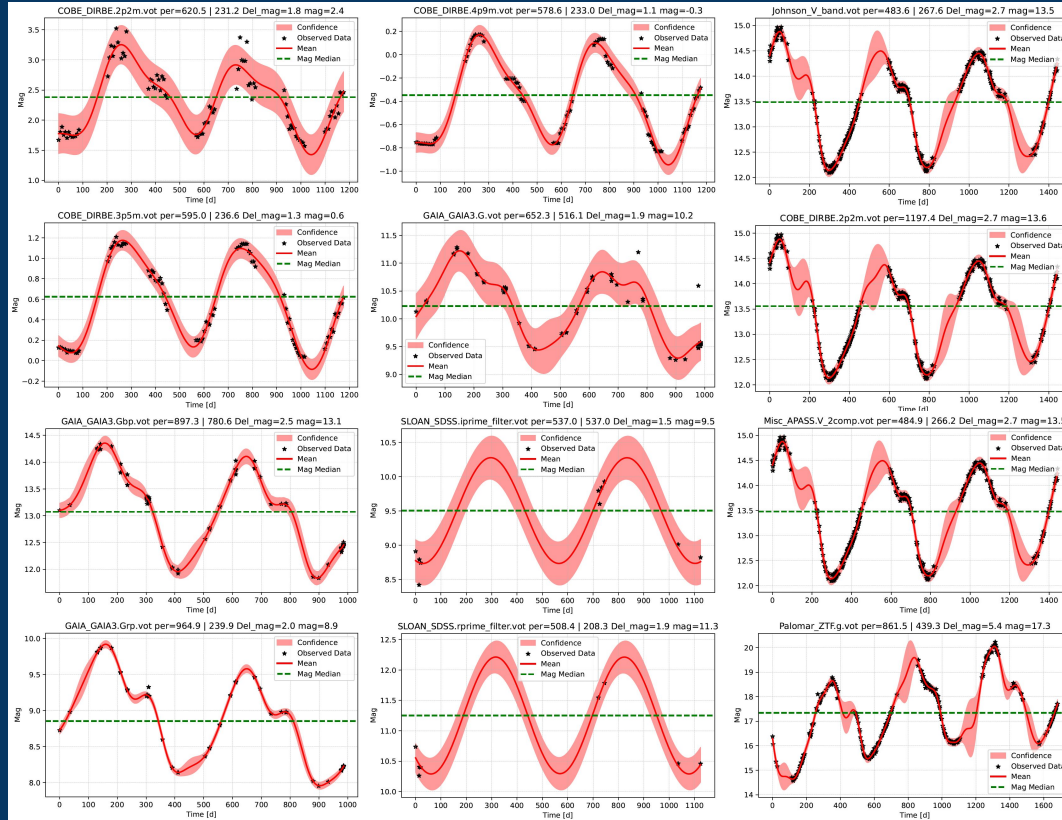
Multiwavelength Variability in Nearby Evolved Stars



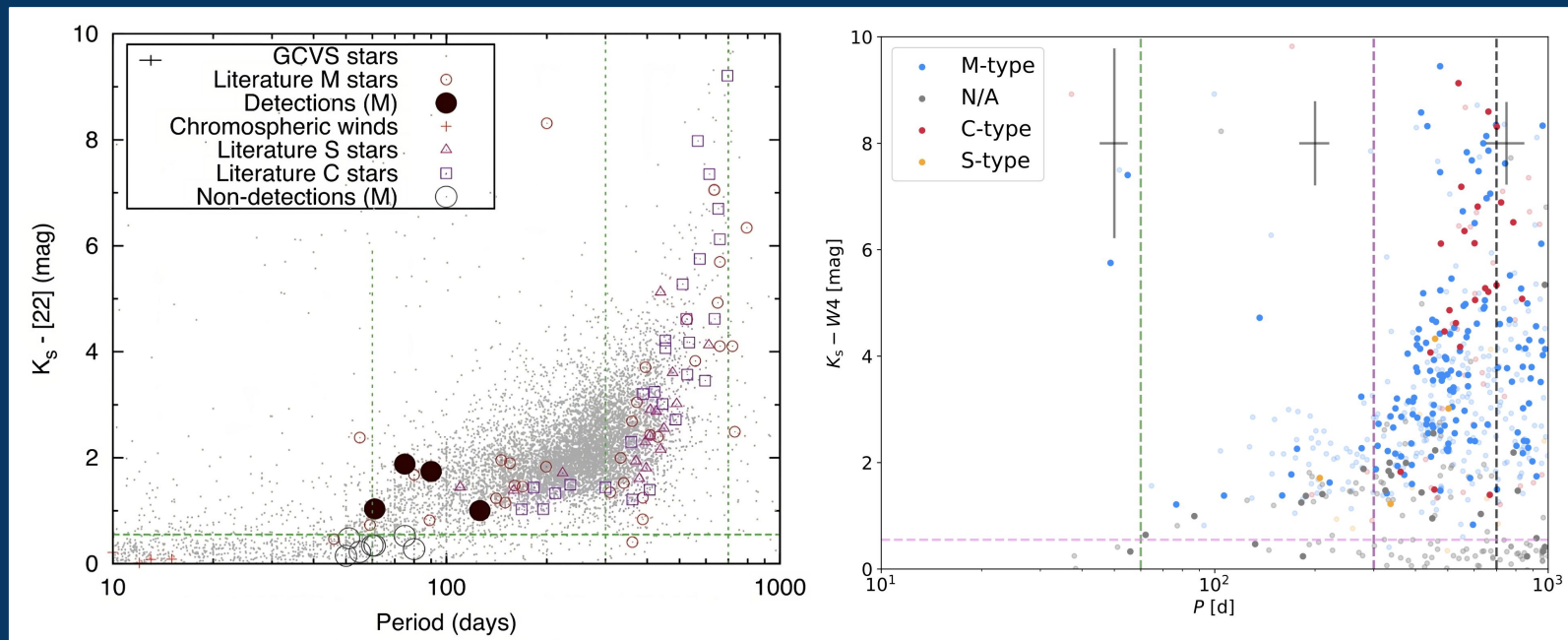
Results

Results: sample fits

IRAS 04307+6210

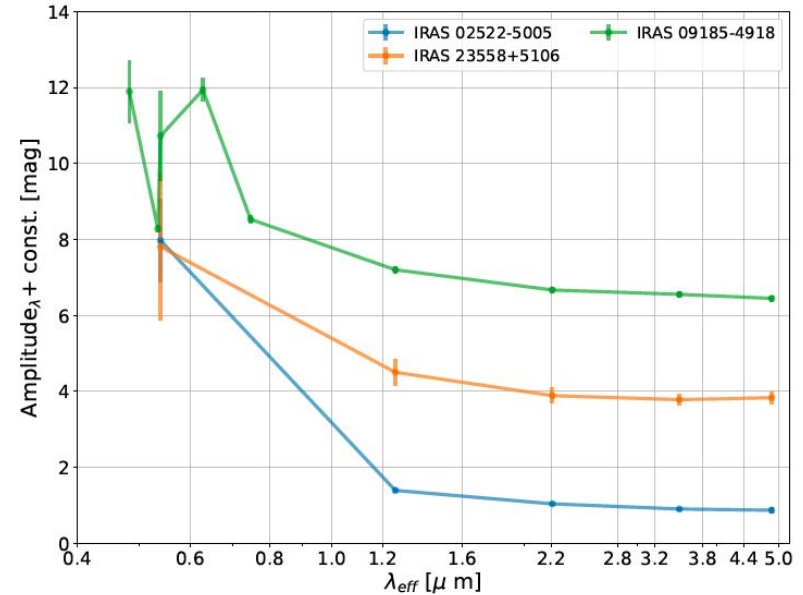
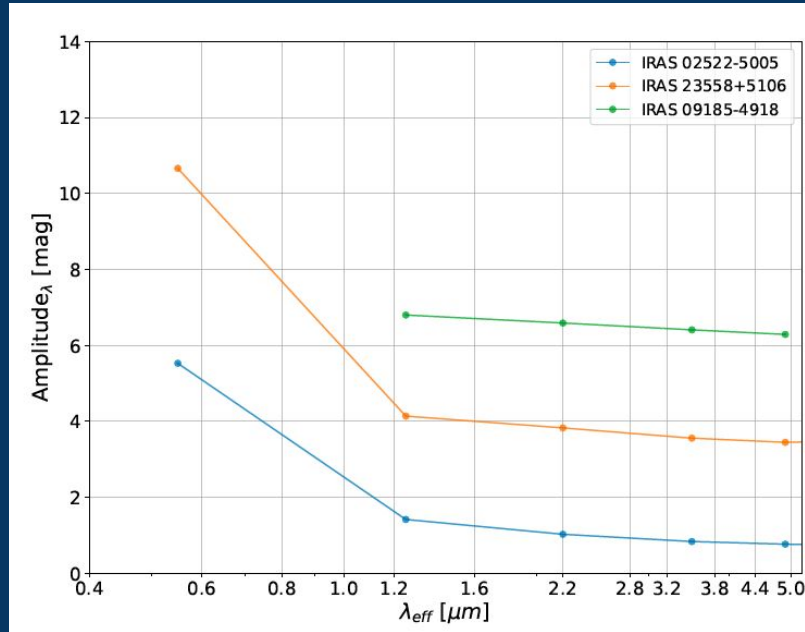


Results: period vs colour

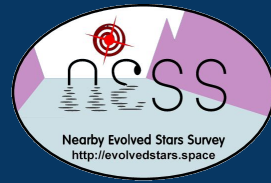


McDonald+ 2018

Results: amplitude vs. wavelength

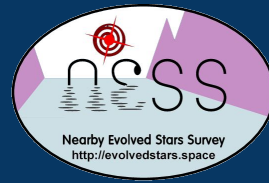


Smith+ 2002



NESS variables: TBD

- PGMUVI + MCMC to investigate parameter degeneracies/correlations.
- PGMUVI multiwavelength fits, SEDs as a function of time.
- P-L relations; compare P-L distances to distances from other methods.
- Physical parameters (T_{eff} , R , L , optical depth) as a function of time.
- Extinction correction.
- Sources with long secondary periods.



Future: Extragalactic Extreme AGB



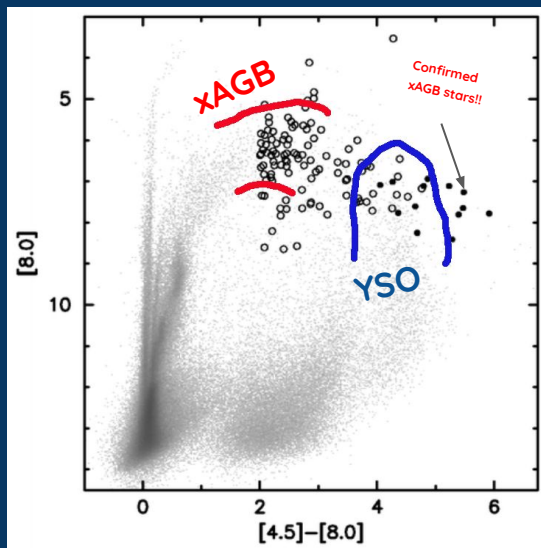
Extragalactic Extreme xAGB

Most evolved ~5% of the AGB population, but contribute ~75% of the dust budget (e.g., Riebel+ 2012, Srinivasan+ 2016).

xAGB stars could be hiding in YSO

samples!!! (e.g., Gruendl+ 2008).

Photometry alone can't separate them.



- Time-series data!
- Magellanic Clouds (e.g., Yang+ 2018, Le+ 2021, Groenewegen 2022).
- Great way to separate **extremely extreme** AGB stars from YSOs (e.g., Groenewegen 2022).



The Nature of the M33 24 μ m variables

THE IDENTIFICATION OF EXTREME ASYMPTOTIC GIANT BRANCH STARS AND RED SUPERGIANTS IN M33 WITH 24 μ m VARIABILITY

Edward J. Montiel¹, Sundar Srinivasan², Geoffrey C. Clayton¹, Charles W. Engelbracht^{3,4}, and Christopher B. Johnson¹

Published 2015 January 15 • © 2015. The American Astronomical Society. All rights reserved.

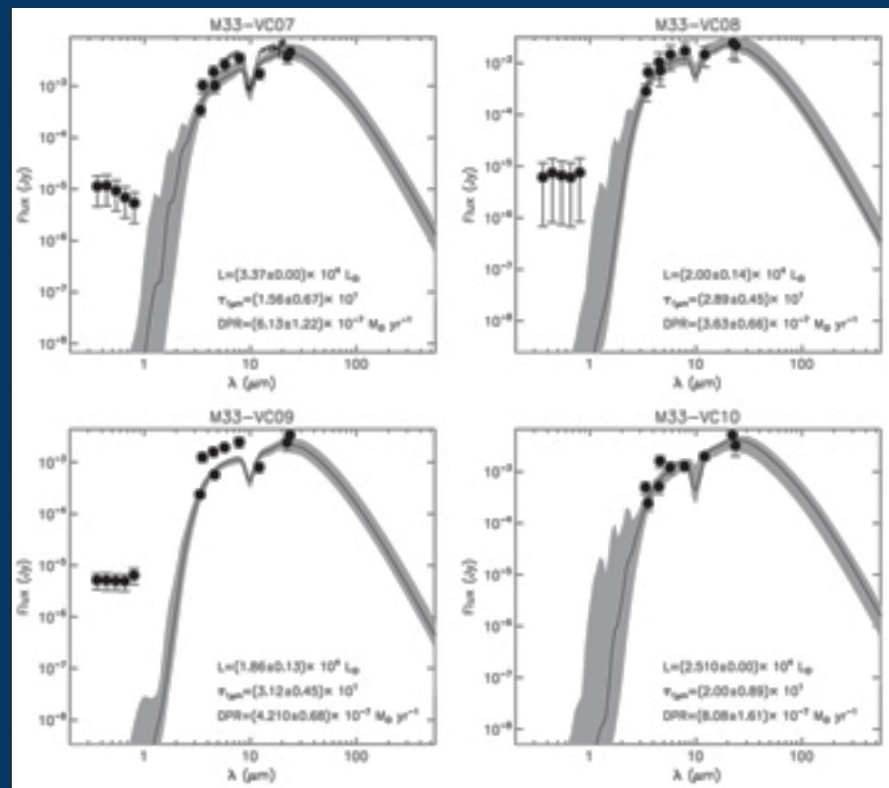
[The Astronomical Journal](#), Volume 149, Number 2

Amplitudes: 10% to 120% (median 50%) of mean 24 μ m flux !

SEDs similar to the most extreme O-rich AGB stars seen in the MWG and Magellanic Clouds. BUT could also be embedded massive YSOs [VC 01].

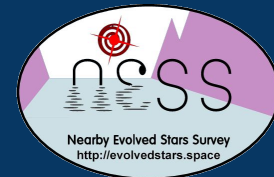
GRAMS (Sargent+ 2011, Srinivasan+ 2011) best-fit models predict VERY high luminosities close to or above the classical AGB limit!

Super-AGB candidates???? Or just YSOs????





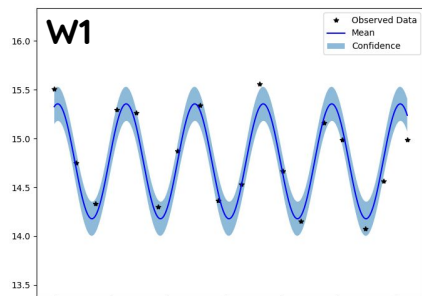
NEOWISE to the rescue!



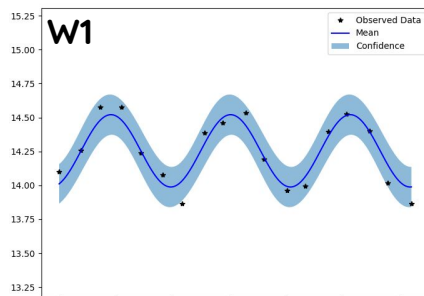
VC 21

VC 23

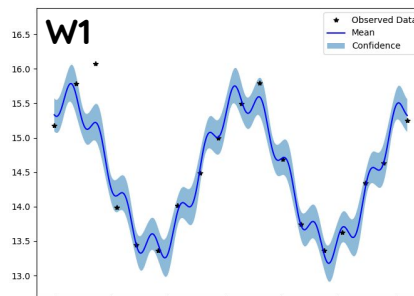
VC 06



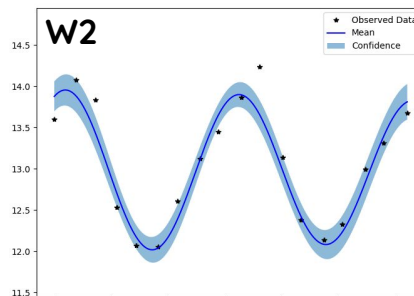
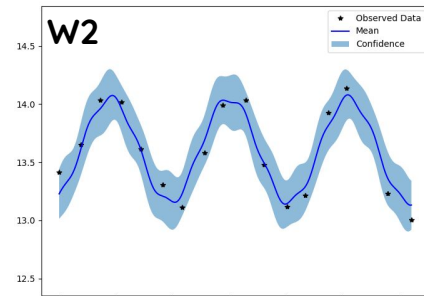
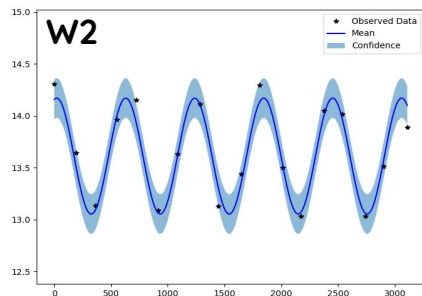
Period ≈ 610 d



Period ≈ 1050 d



Period ≈ 1550 d



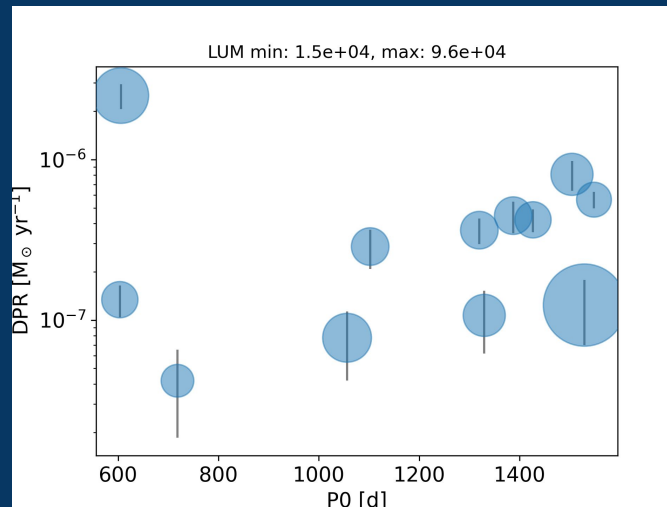
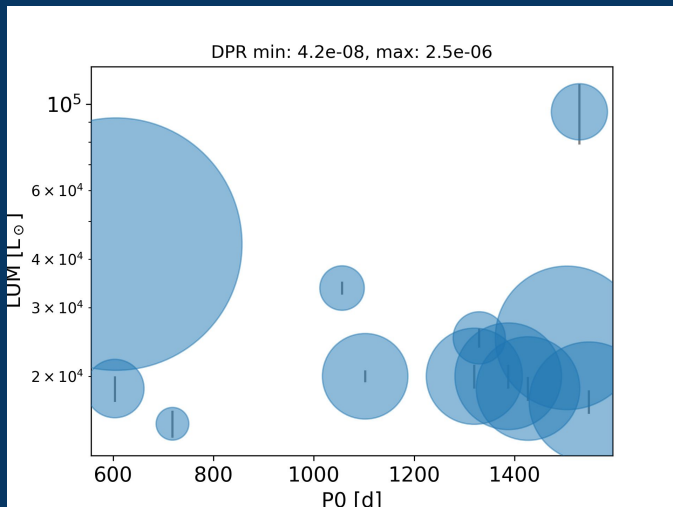
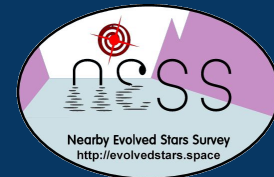
20 of 24 have NEOWISE
light curves in at least
one band.

12 of 20 show simple
sinusoidal variation with
periods between 600
and 1500 d and large
amplitudes!

Definitely evolved stars!



High L, high DPR, high P!



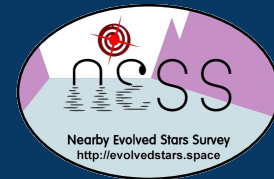
Range in luminosity and
DPR about 1 dex.

No significant trend with
period (small sample).

Probing the very end of
the mass-losing AGB
phase!



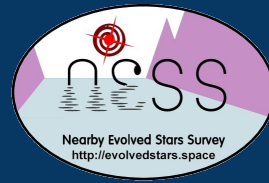
Summary



- Multiwavelength light curves are here! We have code! We can do this!
- >6700 light curves collected for >700 nearby (< 3 kpc) evolved stars!
PGMUVI fits provide periods! Lots of analysis remains to be done!
- Can extend to nearby galaxies! Can exploit rapidly growing multiwavelength database of time series!



Multiwavelength Variability in Nearby Evolved Stars

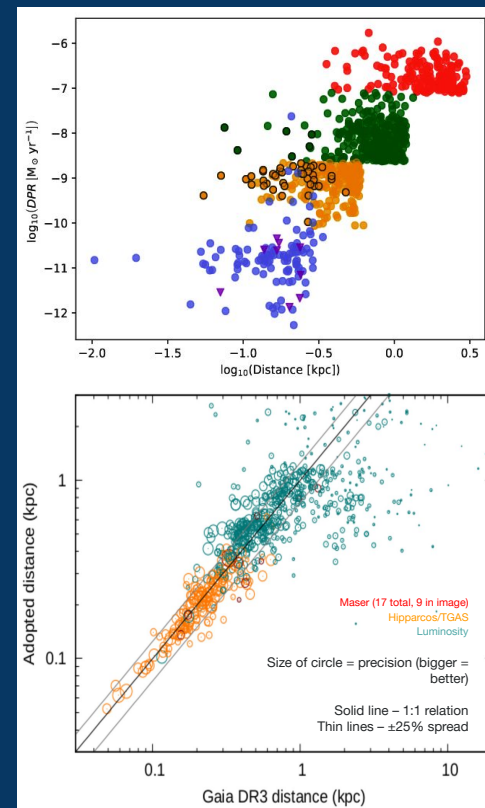


Thanks to you and to the organisers!



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Observations in progress: IRAM, LMT, VLA
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- Lots of archival data over entire relevant range of wavelengths!



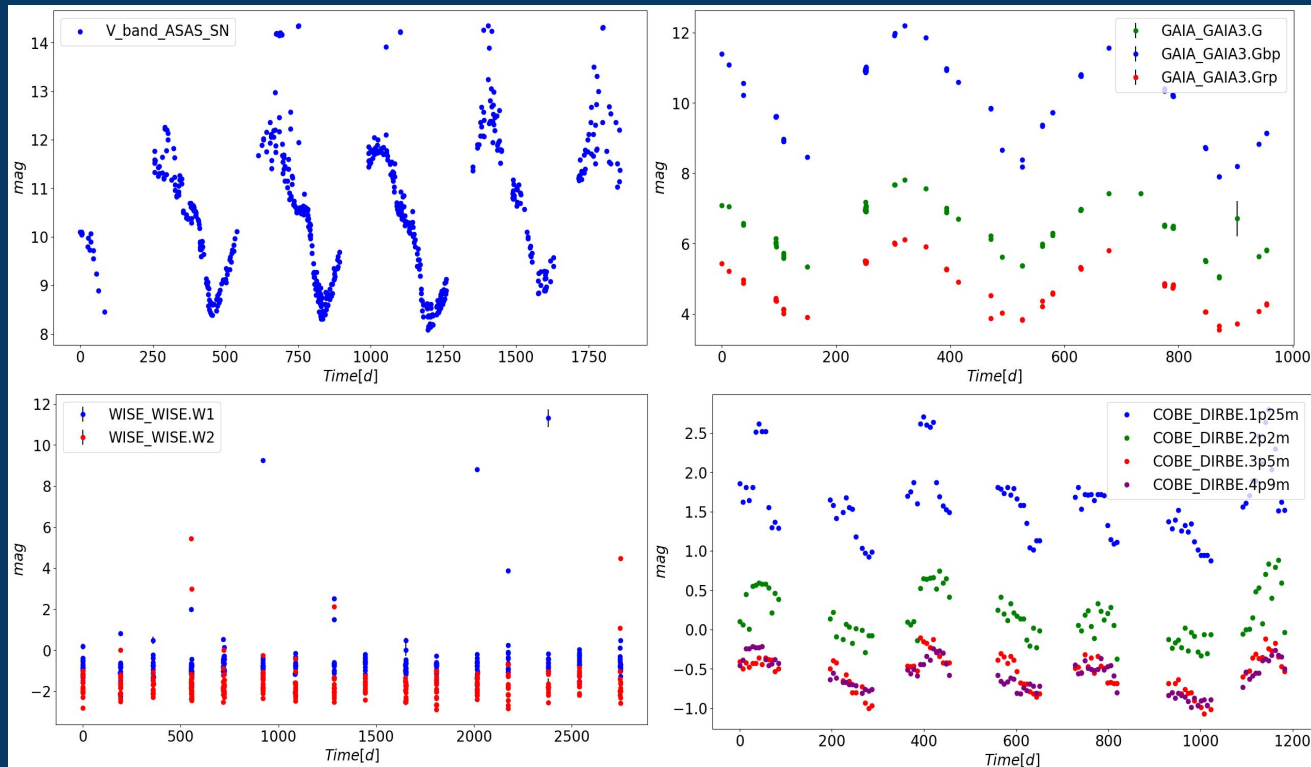


The NESS light-curve sample

ID in this thesis	Spectral Range	Photometric Bands	
		Filter ID [†]	λ_{eff} [nm]
ASAS_SN	Optical	V: Misc_APASS.V	539.3
APASS_DR10	Optical	B: TYCHO_TYCHO.B	428.0
		V: TYCHO_TYCHO.V	534.0
		g': SLOAN_SDSS.gprime_filter	482.0
		r': SLOAN_SDSS.rprime_filter	628.7
		i': SLOAN_SDSS.iprime_filter	745.7
DIRBE	Infrared	z: SLOAN_SDSS.z	899.2
		1.25 μm : COBE_DIRBE.1P25m	1255.4
		2.2 μm : COBE_DIRBE.2P2m	2210.1
		3.5 μm : COBE_DIRBE.3P5m	3491.1
		4.9 μm : COBE_DIRBE.4P9m	4867.4
GAIA_DR3	Optical	BP: GAIA_GAIA3.Gbp	531.9
		G: GAIA_GAIA3.G	671.9
		RP: GAIA_GAIA3.Grp	793.9
KELT	Optical	R_k: OSN_Johnson.Cousins_R3 (_0, _1)	645.5
NEOWISE	Infrared	W1: WISE_WISE.W1	3352.6
		W2: WISE_WISE.W2	4602.8
PANSTARRS	Optical	g: PAN-STARRS_PS1.g	481.0
		r: PAN-STARRS_PS1.r	615.5
		i: PAN-STARRS_PS1.i	750.3
		z: PAN-STARRS_PS1.z	866.8
		y: PAN-STARRS_PS1.y	961.3
WHITELOCK	Infrared	J: CTIO_OSIRIS.J	1258.1
		H: CTIO_OSIRIS.H	1616.2
		K: CTIO_OSIRIS.K	2163.0
		L: IRTF_NSFCAM.L	3472.0
ZTF	Optical	g: Palomar_ZTF.g	472.2
		r: Palomar_ZTF.r	633.8
		i: Palomar_ZTF.i	788.7

- Nine surveys, 29 photometric bands.
- Can study variability as a function of wavelength!
- 712 of 852 targets have a light curve in at least one band.
- Python scripts (`wget` or `ADQL`), will be available with paper.
- Fully reproducible!

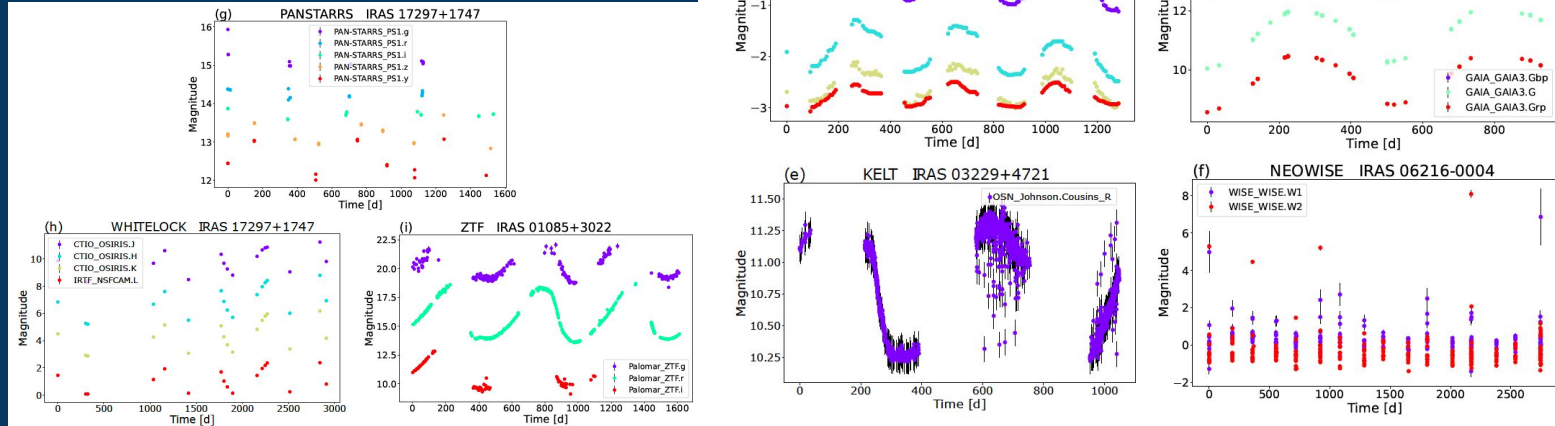
NEOWISE saturation: TBD



Some light curves for T Lep

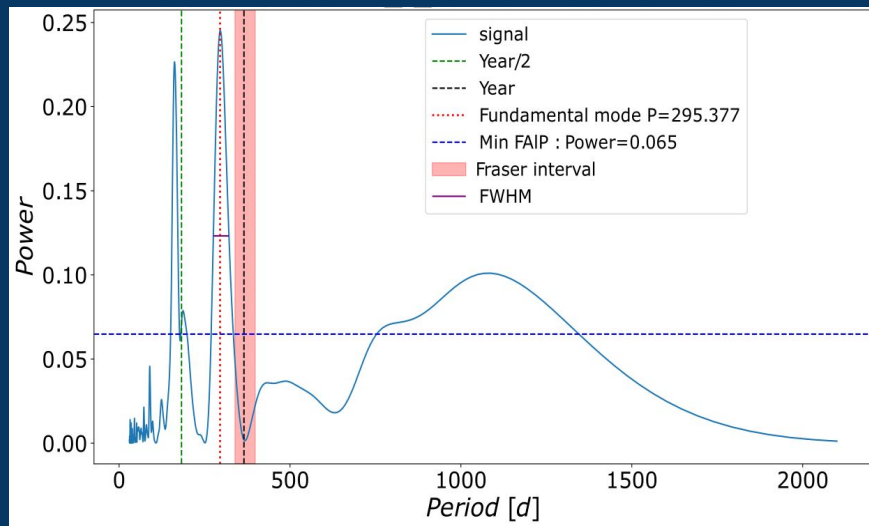
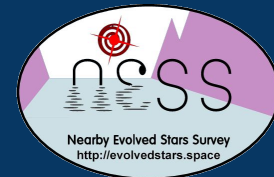
The dataset

- A total of >6700 light curves.
- Fits and data will be publicly available.

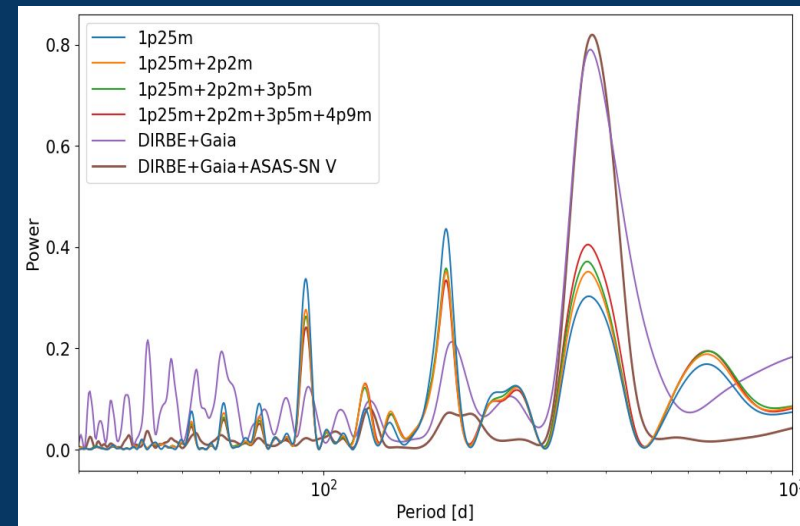




Lomb-Scargle periodograms

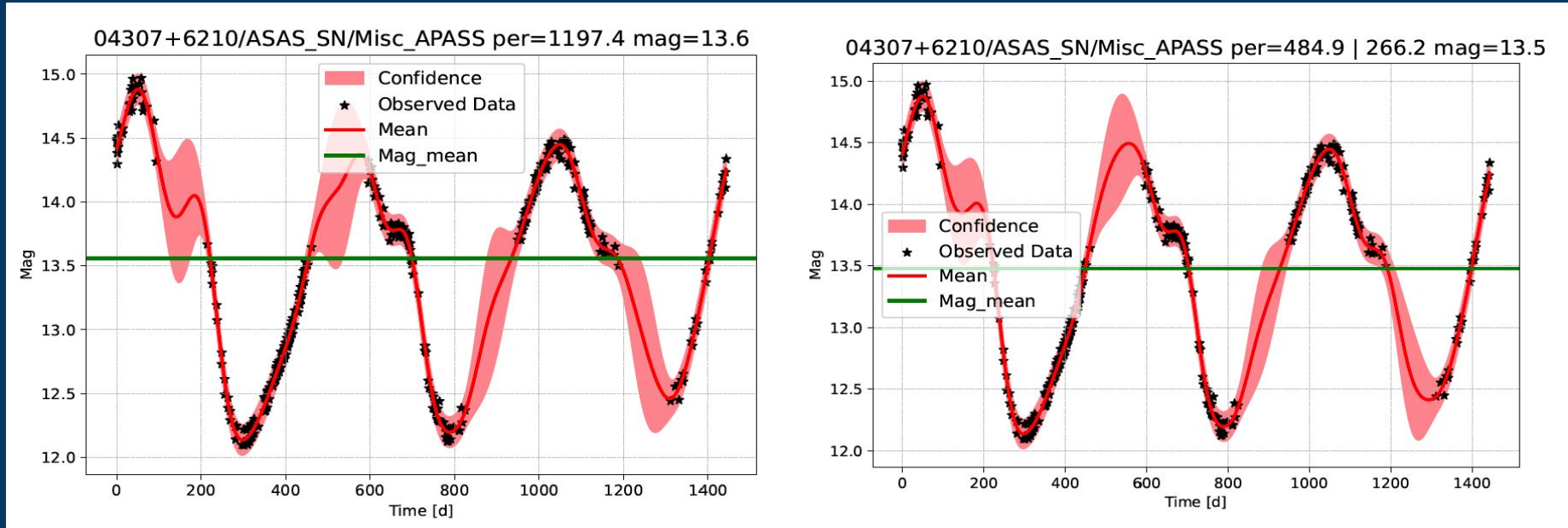


Single-band LS

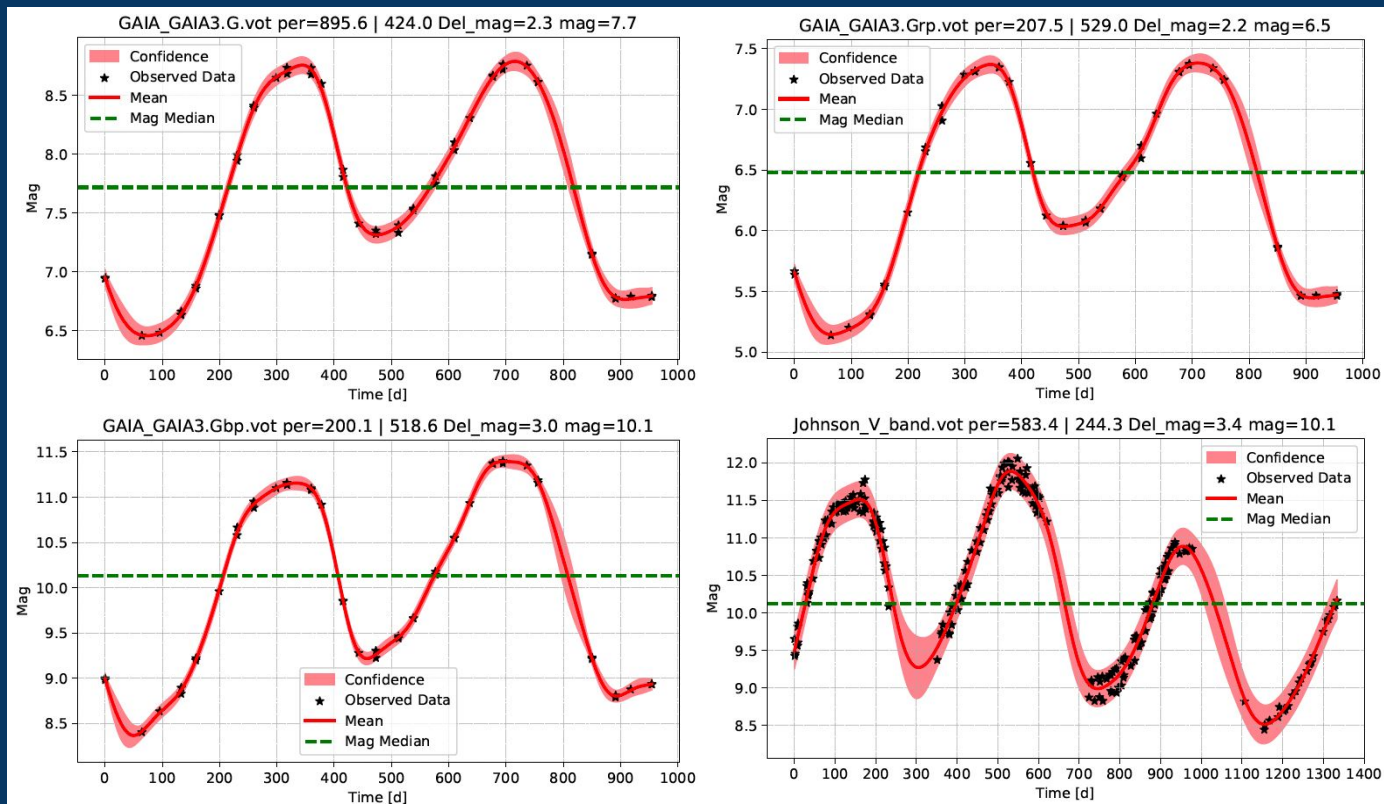


Multi-band LS

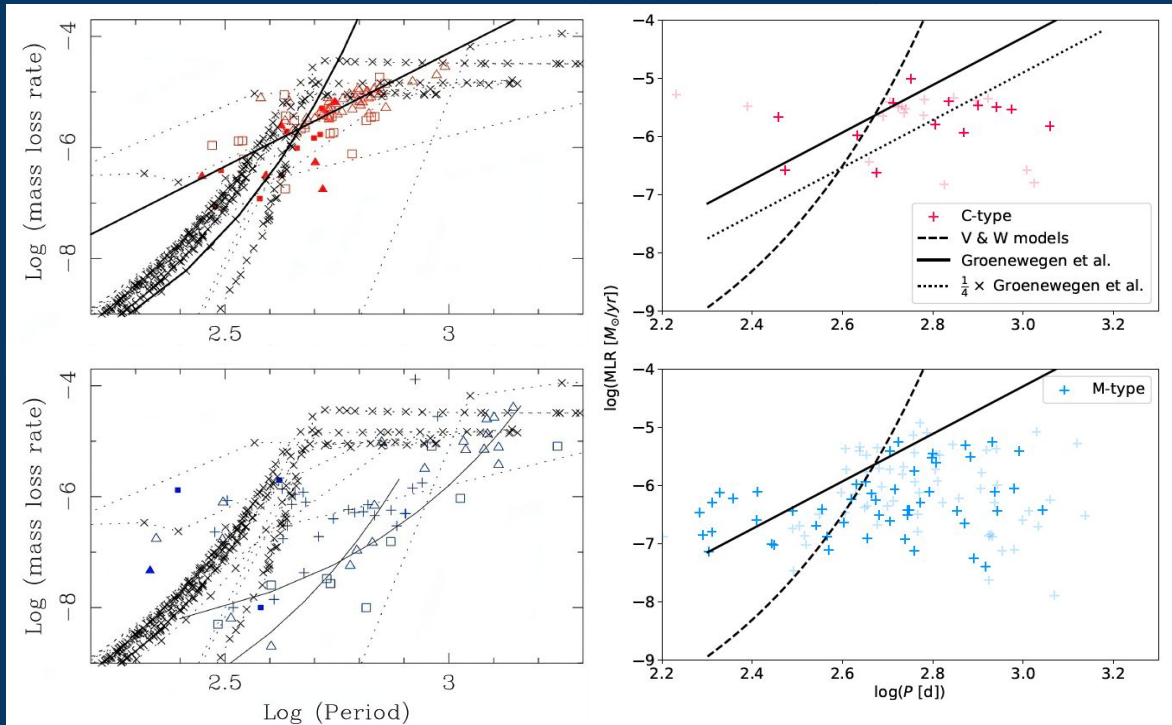
1-comp vs 2-comp fit



Sample fits



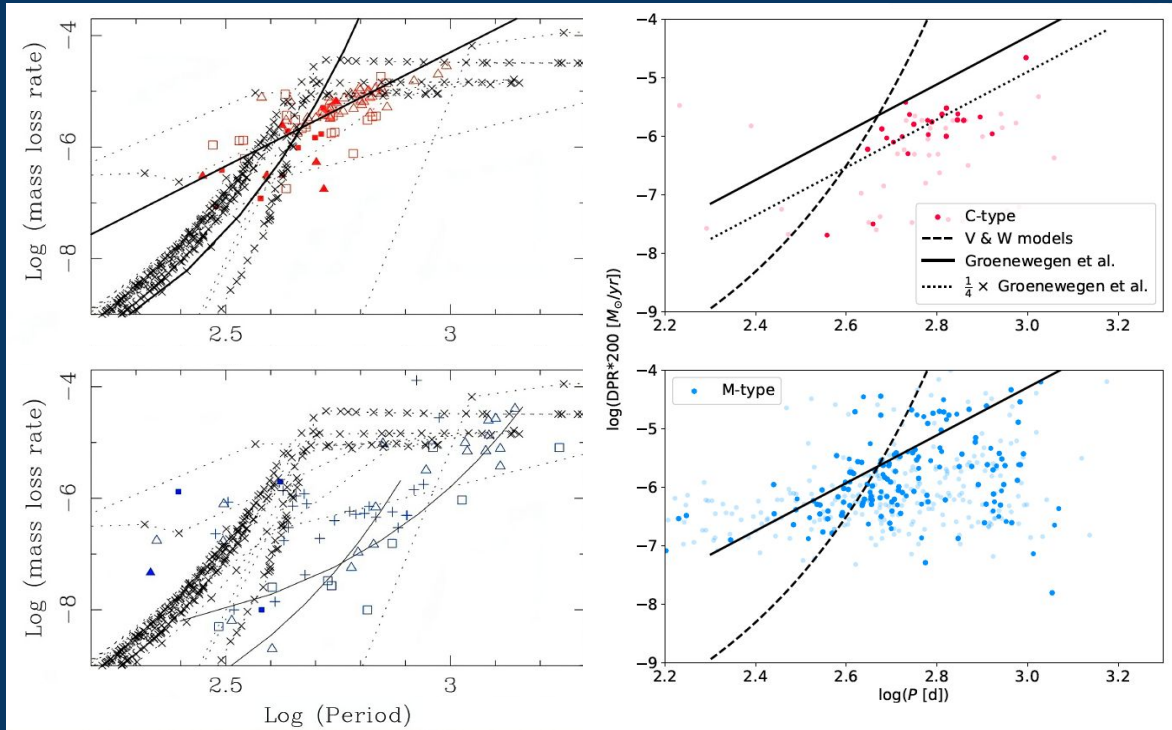
Gas mass-loss rate vs. period



Groenewegen+ 2009

MLRs from Wallström+ in prep

Dust production rate vs. period



Groenewegen+ 2009

DPRs from GRAMS (Sargent+ 2011, Srinivasan+ 2011) fits
(Trejo+ in prep)

Sample fits

